Grade 9 – Mathematics

Third Term Model Paper 2

Chalaka Perera | BSc (Hons) Computer Engineering, University of Peradeniya

Name / Index No.: ..... **Time: 3 hours**

- This paper consists of two parts: Part I and Part II.
- Part I: Answer **all** questions on this paper itself. Part A has 15 questions (2 marks each). Part B has 2 questions (10 marks each).
- Part II: Answer **five** questions only, on separate sheets. Each question carries 10 marks.
- Show all relevant steps of working. Write units where necessary. Use π = 22/7 where needed.

For examiner's use only	
PART I – A	/ 30
PART I – B	/ 20
PART II	/ 50
Total	**/ 100**

PART I

Part A – Answer all questions in the space provided. (2 marks each)

1. Find the sum: $1101_2 + 111_2$ (give the answer in base two) (2)

Answer: .....

2. Find the value: $(2^3)^2 \times 2^{-4}$ (2)

Answer: .....

3. Round off 3.4672 to two decimal places. (2)

Answer: .....

4. The nth term of a number pattern is 4n – 1. Find its 20th term. (2)

Answer: .....

5. ¾ of a number is 27. Find the number. (2)

Answer: .....

6. 18% of a sum of money is Rs 54. Find the sum of money. (2)

Answer: .....

7. Expand and simplify: (x + 5)(x – 3) (2)

Answer: .....

8. A rectangular tank is 2 m long, 1.5 m wide and 1 m high. How many litres of water can it hold when full? (2)

Answer: .....

9. A and B are two fixed points. Name the locus of the points which are equidistant from A and B. (2)

Answer: .....

10. Using the formula s = (a + b + c)/2, find s when a = 5, b = 7 and c = 8. (2)

Answer: .....

11. Show that a triangle with sides 7 cm, 24 cm and 25 cm is a right-angled triangle. (2)

Answer: .....

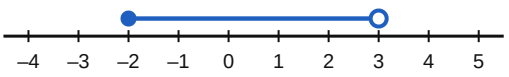
12. The mass of an object is 3.2×10^5 g. Write this as an ordinary number and in kilograms. (2)

Answer: .....

13. Find the magnitude of an exterior angle of a regular octagon. (2)

Answer: .....

14. Write the inequality represented on the number line, and write all the integers that satisfy it. (2)



Answer:

.....

15. Simplify: 2/x + 3/(2x) (2)

Answer: .....

Part B – Answer both questions on this paper. (10 marks each)

1.

A ship leaves a harbour H and sails 40 km on a bearing of 060° to a point A. From A it sails 30 km on a bearing of 150° to a point B.

(i) Draw a rough sketch showing this information. (2 marks)

(ii) Show that HÂB = 90°. (2 marks)

(iii) Using Pythagoras' relation, find the distance HB. (2 marks)

(iv) Using the scale 1 cm = 10 km, draw a scale diagram. (2 marks)

(v) Using your scale diagram, find the bearing of B from H. (2 marks)

2.

A rectangular water tank is 1.2 m long, 0.8 m wide and 1 m high.

(i) Find the volume of the tank in cubic metres. (2 marks)

(ii) Find the capacity of the tank in litres. (2 marks)

(iii) Water flows into the empty tank at 40 litres per minute. How long does it take to fill the tank? (2 marks)

(iv) A family uses 240 litres of water per day. For how many days will a full tank last? (2 marks)

(v) Find the depth of water in the tank when it contains 480 litres. (2 marks)

PART II

Answer five questions only. (10 marks each)

1.

(a) A man spends 2/5 of his monthly salary on food and 1/4 of the remainder on rent. He saves the rest.

(i) What fraction of his salary is spent on rent? (2 marks)

(ii) What fraction of his salary does he save? (2 marks)

(iii) If he saves Rs 27 000, find his monthly salary and the amount spent on food. (3 marks)

(b) A land agent receives a commission of 3% on the value of the land he sells. Find his commission when he sells a land for Rs 2 500 000. (3 marks)

2.

Incomplete tables of values for the functions $y = 3 – x$ and $y = x + 1$ are given below.

$y = 3 – x$

x	–1	0	1	2	3	4
y	4	…	2	…	0	–1

$y = x + 1$

x	–1	0	1	2	3	4
y	0	1	…	3	…	5

(i) Fill in the blanks in both tables. (2 marks)

(ii) Draw the graphs of both functions on the same coordinate plane. (4 marks)

(iii) Write the gradient of each graph. (2 marks)

(iv) Write the coordinates of the point of intersection of the two graphs. (2 marks)

3.

(a) Factorise: (i) $x^2 – 9x + 20$ (ii) $9a^2 – 16b^2$ (iii) $3x^2 + 6x$ (5 marks)

(b) Simplify: $(x + 1)/2 – (x – 3)/5$ (3 marks)

(c) Using factors, find the value of $103^2 – 97^2$. (2 marks)

4.

(a) Solve: $5(x – 2) – 3(x + 1) = 7$ (3 marks)

(b) The sum of two numbers is 25 and their difference is 7. Form a pair of simultaneous equations and find the two numbers. (4 marks)

(c) The length of a rectangle is 4 cm more than its breadth, and its perimeter is 36 cm. Find its area. (3 marks)

5.

Use only a straight edge with a cm/mm scale and a pair of compasses for the following constructions. Show all construction lines.

(i) Draw a straight line segment AB of length 7 cm. (1 marks)

(ii) Construct an angle of 60° at A such that AB is an arm. (2 marks)

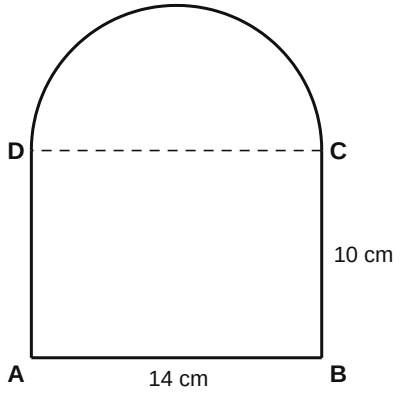
(iii) Construct an angle of 45° at B such that BA is an arm, on the same side as the angle at A. Name the point where the arms meet as C. (3 marks)

(iv) Measure and write the magnitude of $A\hat{C}B$ and the length of AC. (2 marks)

(v) Construct the bisector of $A\hat{C}B$. (2 marks)

6.

The figure is made of a rectangle ABCD with a semicircle drawn on the side DC. AB = 14 cm and BC = 10 cm. ($\pi = 22/7$)



(i) Find the area of the rectangle. (2 marks)

(ii) Find the area of the semicircle. (2 marks)

(iii) Find the total area of the figure. (1 marks)

(iv) Find the perimeter of the figure. (3 marks)

(v) A square has the same perimeter as this figure. Find the area of the square. (2 marks)

7.

(a) A fair spinner is divided into 8 equal sectors numbered 1 to 8. It is spun once. Find the probability that the number obtained is

(i) an even number (ii) a prime number (iii) a multiple of 3 (4 marks)

(b) The marks obtained by 10 students for a test are: 45, 52, 60, 38, 52, 70, 65, 52, 41, 55. Find the

(i) mode (ii) median (iii) mean (iv) range (6 marks)